

Note to junior batches:

Pleaseeee solve past papers of Dr. Kamran and Dr. Faheem!!

For Reference:

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
TSCS, Midterm Examinations Spring 2022

Time: 90 minutes **Theory of Programming Language (CT-367) - A** **Max Marks: 20**

Note: Attempt all questions.

Q-1) Explain factors, which influence the design of any programming language. Also state general attributes of a good programming language? (4)

Q-2) What are the models of computation? Explain with suitable examples. (4)

Q-3) Draw and explain the compilation process life cycle? (4)

Q-4) Differentiate between ambiguous and unambiguous grammar? (4)

Q-5) Write lex program of any one of the following: (4)

- a) Validate mobile number of any of four telecom operators?
- b) Check a valid URL?

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0300 J02
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DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
TSCS, Midterm Examinations Spring 2022

Time: 90 minutes **Theory of Programming Language (CT-367) - B** **Max Marks: 20**

Note: Attempt all questions.

Q-1) What are different programming domains? Briefly explain the language evaluation criteria? (4) (C-2)

Q-2) Briefly explain the process of lexical analysis? (4) (C-2)

Q-3) What is lambda calculus? Illustrate its computation usage with examples? (4) (C-2)

Q-4) Differentiate between Julia and other types programming languages (Python etc.)? (4) (C-3)

Q-5) Write lex program of any one of the following: (4) (C-3)

- a) Validate date format (Day\Month\Year)?
- b) Check either an expression is mathematical or relational operator based?

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Note: Attempt all questions.

Q-1) Explain factors, which influence the design of any programming language. Also state general attributes of a good programming language? (4) (C-2)

Q-2) What are the models of computation? Explain with suitable examples. (4) (C-2)

Q-3) Draw and explain the compilation process life cycle? (4) (C-2)

Q-4) Differentiate between ambiguous and unambiguous grammar? (4) (C-3)

Q-5) Write lex program of any one of the following: (4) C-3

- a) Validate mobile number of any of four telecom operators?
- b) Check whether given string is Palindrome or Not?

TSCS, Midterm Examinations Spring 2023

Time: 90 minutes Theory of Programming Language (CT-367) - B Max Marks: 20

Note: Attempt all questions.

Q-1) What are different programming domains? Briefly explain the language evaluation criteria? (4)

Q-2) Briefly explain the process of lexical analysis? (4)

Q-3) What is lambda calculus? Illustrate its computation usage with examples? (4)

Q-4) Differentiate between flex and bison? (4)

Q-5) Write lex program of any one of the following: (4)

- a) Validate date format (Day\Month\Year)?
- b) Check either an expression is mathematical or relational operator based?

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
TSCS, Midterm Examinations Spring 2026

Time: 90 minutes

Theory of Programming Language (CT-367) - A

Max Marks: 20

Note: Attempt all questions.

- Q-1) Explain factors, which influence the design of any programming language. Also state general attributes of a good programming language? (3) (CLO-1)
- Q-2) What are the models of computation? Explain with suitable examples. (3) (CLO-1)
- Q-3) What is currying operation? How to model **numeric increment** in Lambda Calculus with an example? (CLO-2)
- Q-4) Differentiate between following: a) Ambiguous vs Unambiguous grammar b) Deterministic vs Non-Deterministic grammar? (4) (CLO-2)
- Q-5) Write flex or bison program of the following: a) Validate PTCL number of any three cities of Pakistan? b) Check whether given string is Palindrome or Not? (6) (CLO-2)

Name: Nashrey

Seat # 32

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR (B.S IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2023
Batch 2020

Time : 3 Hours

Dated : 07-AUG-23
Max Marks : 60

Theory of Programming Languages - CT-367

Note: Attempt all questions. All questions carry equal marks.

- ☒ Q.1 (a) State the names of the programming paradigms and briefly explain their main features. Compare the Imperative language over the Declarative language with an example. (C-2) 5
- ☒ (b) Analyse and explain the process of lexical analysis. Elaborate how regular expressions are utilized to make lexemes in the compilation process with appropriate examples. (C-2) 5
- ☒ Q.2 (a) Explain the process of semantics analysis during compilation cycle of a source code of a programming language. Also, discuss the various categories of semantics. (C-2) 5
- ☒ (b) Ambiguous expression will be converted into an unambiguous expression! Explain with an example. (C-2) 5
 $E \rightarrow E + E / E * E / \dots$ $E \rightarrow E - P / P$
 $P \rightarrow P * Q / Q$
- ☒ Q.3 (a) For subprograms or functions, what are the design issues? Also, list down categories of subprograms. (C-3) 5
- ☒ (b) Elaborate and discuss the categories of variables by lifetime with advantages and disadvantages. (C-3) 5
- ☒ Q.4 (a) Referential transparency exposes the functional side effects! Discuss both the concepts with examples. (C-3) 5
- ☒ (b) Order of operand evaluation is important in expression evaluation! Discuss the rules associated with it. Also explain type conversion. (C-3) 5
- ☒ Q.5 (a) Discuss, the concept of type checking. Differentiate between static type checking and dynamic type checking and give their relative advantages. (C-3) 5
- ☒ (b) Initiate a source code with the help of Flex and Bison tools that accepts all of these expressions given below: (Note: provide files with extension .lex and .y) (C-3) 5
- > (identifier * identifier)
 - > (identifier * constant)
 - > (identifier / identifier)
 - > (constant / identifier)
- ☒ Q.6 (a) What are Top-up and Bottom-up parsing techniques? How PDA will help to implement Top-up parsing technique, discuss with an appropriate example. (C-3) 5
- ☒ (b) Briefly explain the concept of short-circuit evaluation with help of an example. Also, list down the languages which utilizes this concept of expression evaluation. (C-3) 5

---- BEST OF LUCK ----

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR (Computer Science)
SPRING SEMESTER EXAMINATIONS 2025

Batch 2022

Dated : 04-JUN-25
Max Marks : 60

Time : 3 Hours

Sir Jasim

Theory of Programming Languages - CT-367

Note: Attempt all questions. All questions carry equal marks.

- Q.1 (a)** State the names of the programming paradigms and briefly explain their main features. Compare the Imperative language over the Declarative language with an example. (C-2) 5
- (b)** Analyse and explain the process of Syntax analysis. Elaborate how context free grammar is utilized to make parse trees in the compilation process with appropriate examples. (C-2) 5
- Q.2 (a)** Briefly explain the compilation process of a source code of a programming language. Also, draw a labelled diagram. (C-2) 5
- (b)** Ambiguity in expressions will have an effect on the compilation process! Explain with an example. Also discuss the recommended solution. (C-2) 5
- Q.3 (a)** Explain the difference between static scope and dynamic scope. Support your explanation with an example in pseudo code or a language you are familiar with. Indicate what output would be produced under each type of scope. (C-3) 5
- (b)** Consider the following C code: (C-3)
- ```
#include <stdio.h>
void display() { printf("Display from base-function\n");}
int main() { void (*func_ptr)() = display; func_ptr(); return 0; }
```
- i) What kind of binding is used for the function call func\_ptr()? 5  
ii) Explain how the compiler determines the function to be called.
- Q.4 (a)** Referential transparency exposes the functional side effects! Discuss both the concepts with practical examples. (C-3) 5
- (b)** Differentiate between unary, binary and ternary operators with syntax examples. Can the ternary operator be nested? Write a short example. (C-3) 5
- Q.5 (a)** Discuss the concept of short-circuit evaluation Order of operand evaluation is important in expression evaluation! Discuss the rules associated with it. Also explain type conversion. (C-3) 5
- (b)** Initiate a source code with the help of Flex and Bison tools that accepts all of these expressions given below: (Note: provide files with extension .lex and .y) (C-3) 5
- a) (identifier + identifier) b) (identifier - constant)  
c) (identifier \* identifier) d) (identifier / constant)
- Q.6 (a)** Define static binding and dynamic binding. Explain the difference between them with suitable examples. Which one is used in the C programming language for function calls and why? (C-3) 5
- (b)** What is short-circuit evaluation in C? Explain how it works with the logical operators '&&' and '||'. Why it is useful in conditional expressions? (C-3) 5